HIGH PERFORMANCE COMPUTING FOR SCIENCE AND ENGINEERING

Harvard University
CS 2050

Lecture 19

April 7, 2026

EXAMPLE 1

GAUSSIAN PROCESSES AND POETRY

(final example from Week 9)

QUIZ 9

COMPSCI 2050 > Quizzes > Quiz 9

View as Student

2025-2026 Spring

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Quiz 9

The quizzes are designed to take no more than 30 minutes, but one hour will be allowed.

Do not click into the quiz until you are ready to take it.

Quiz Type	Graded Quiz
Points	8
Assignment Group	Quizzes
Shuffle Answers	Yes
Time Limit	60 Minutes
Multiple Attempts	No
View Responses	Always
Show Correct Answers	Immediately
One Question at a Time	No

Spring Recess

Week 8 - Extended Example: Training Large Language Models

Week 9 - Extended Example: Distributed Linear Algebra

Week 10 - Extended Example: Molecular Dynamics

Week 11 - Extra Topic: Kokkos

Week 12 - Project Work Week

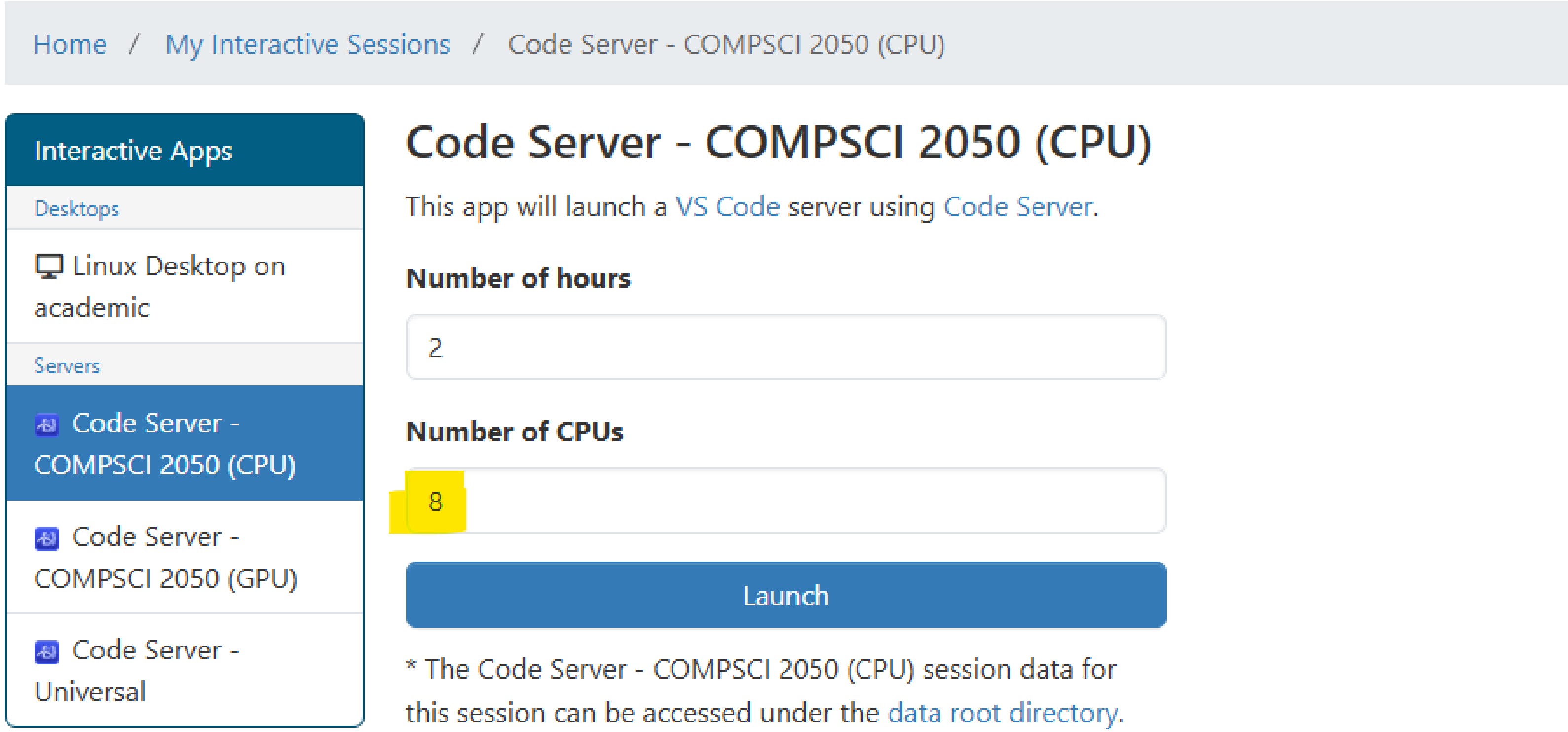
Week 13 - Review

EXAMPLE SETUP #1

- Pull the latest Lecture Examples
- Go to Example 2 and procure a compute node

EXAMPLE SETUP #2

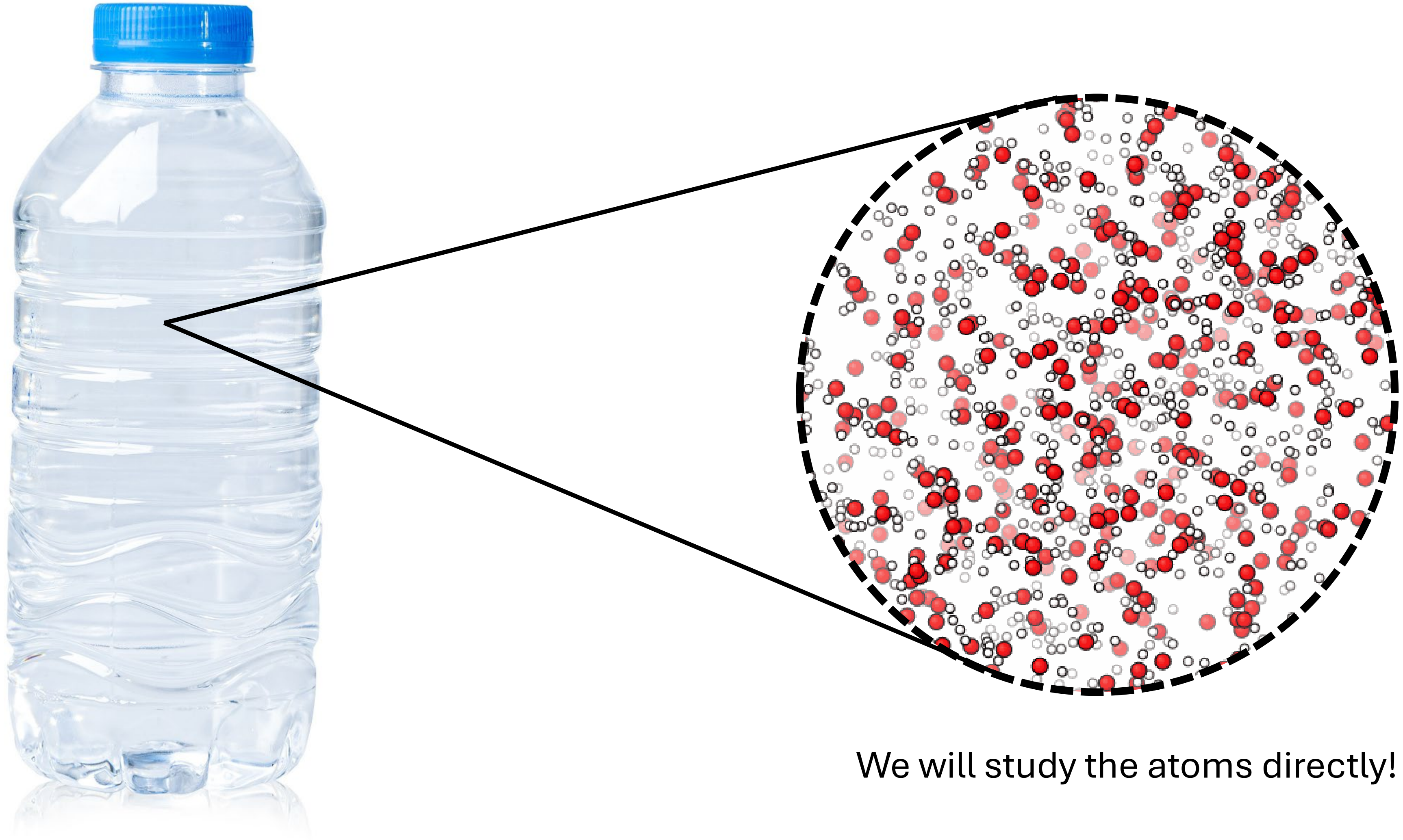
- Launch VS Code



MOLECULAR DYNAMICS SIMULATION

Introduction

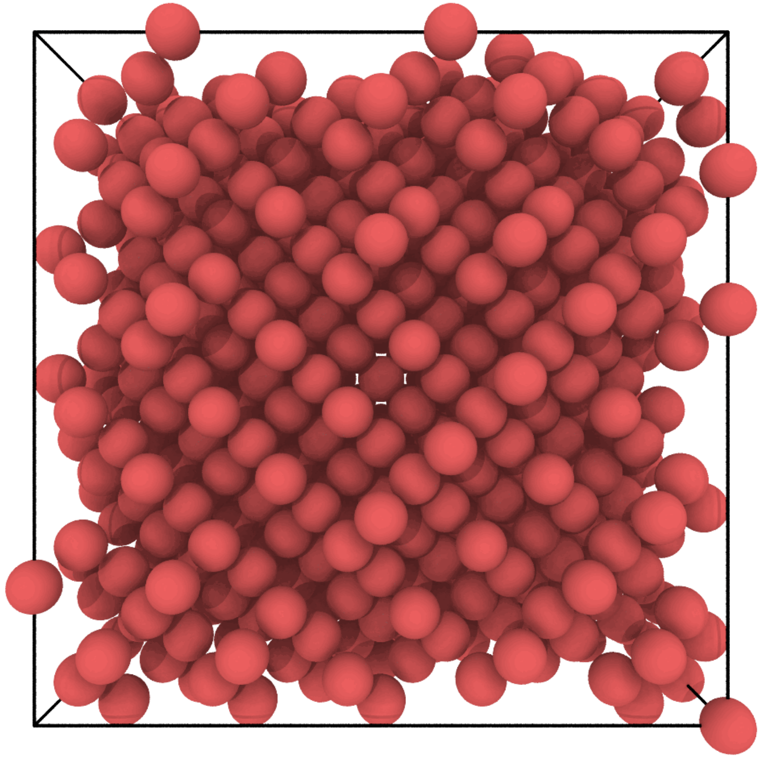
HOW DO WE UNDERSTAND WHY?



We will study the atoms directly!

ATOMISTIC INSIGHTS ARE THE GOAL OF MOLECULAR DYNAMICS

- This week we will simulate the melting of water



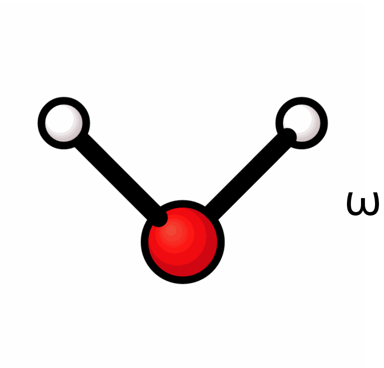
WHY IS THIS HIGH-PERFORMANCE COMPUTING?

- At the resolution of atoms, the world is really, really big.

Spatial Scale

500 mL of water contains ~10^{25} atoms

Temporal Scale



$\omega \approx 10^{-14}$ seconds

To resolve an O-H bond vibration, we need > 5 fs sampling rate (Nyquist Frequency)

An exascale computer would run for >10^{32} years to simulate 1 second of a water bottle.

TO MAKE PROGRESS, WE NEED SMART PHYSICS AND ALGORITHMIC CHOICES

- Atomic systems are complicated
  - We need to make good approximations

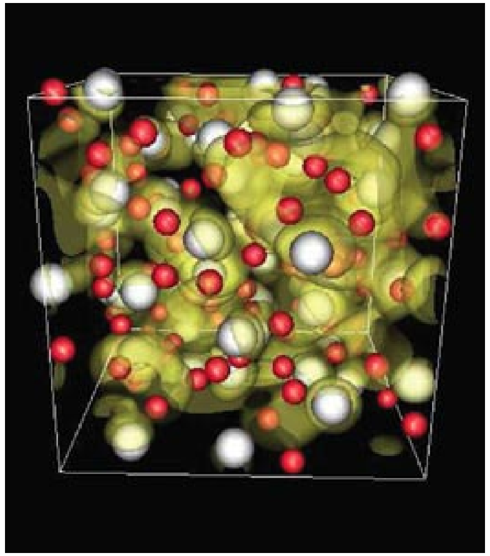
- We need an efficient and parallelizable implementation
  - Anything we want to simulate is going to require many repeated operations

MOLECULAR DYNAMICS SIMULATION

Algorithms and Physics

HOW DO WE REPRESENT OUR SYSTEM?

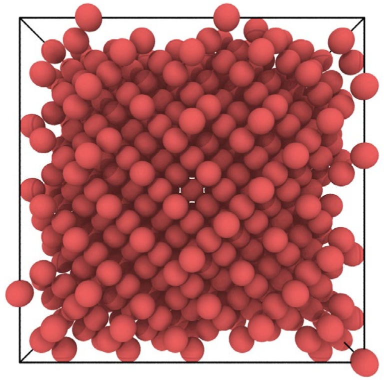
- Molecules and materials at the atomic scale are quantum mechanical systems

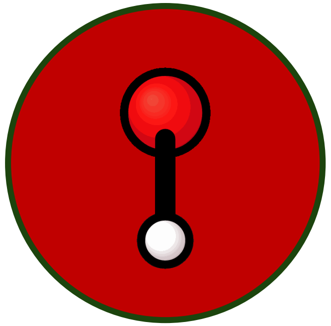


https://cerncourier.com/a/metallic-water-becomes-even-more-accessible/

  - While one could perform simulations with electronic resolution, this is too expensive for our goals: $O(N^3)$ + large pre-factor

- We approximate that atom nuclei will act like classical objects





CLASSICAL SYSTEMS FOLLOW NEWTON'S EQUATIONS

$$E_{\text{total}} = E_{\text{kinetic}}(\mathbf{p}) + E_{\text{potential}}(\mathbf{r})$$

$$\mathbf{F} = m\mathbf{a}$$

- We will numerically integrate this equation:

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\,\Delta t + \frac{1}{2}\mathbf{a}(t)\,\Delta t^2$$

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \frac{1}{2}\left(\mathbf{a}(t) + \mathbf{a}(t + \Delta t)\right)\Delta t$$

CLASSICAL SYSTEMS FOLLOW NEWTON'S EQUATIONS

- The velocity-Verlet Algorithm is our MDEngine's core

Algorithm 1 Velocity Verlet Integration

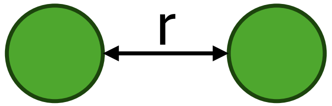
for each timestep **do**
$\quad \mathbf{v}\left(t+\frac{\Delta t}{2}\right) \leftarrow \mathbf{v}(t) + \mathbf{a}(t)\frac{\Delta t}{2}$
$\quad \mathbf{r}(t+\Delta t) \leftarrow \mathbf{r}(t) + \mathbf{v}\left(t+\frac{\Delta t}{2}\right)\Delta t$
$\quad \mathbf{a}(t+\Delta t) \leftarrow \frac{1}{m}\mathbf{F}(\mathbf{r}(t+\Delta t))$
$\quad \mathbf{v}(t+\Delta t) \leftarrow \mathbf{v}\left(t+\frac{\Delta t}{2}\right) + \mathbf{a}(t+\Delta t)\frac{\Delta t}{2}$
end for

- We need to define $E_{\text{potential}}(\mathbf{r})$ and its gradient

FORCES AND ATOMIC INTERACTIONS

- Two atoms interact with each other with a combination of attraction and repulsion

Stillinger-Weber Pair Potential



At r_0 two atoms feel no force as $-\nabla E = 0$

$r = r_0$

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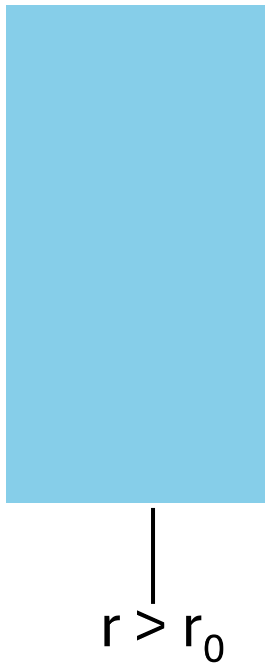
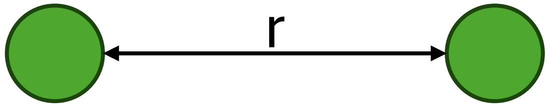
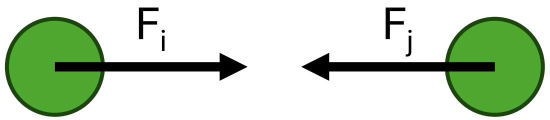




A restoring force will pull the two atoms together towards the energy minimum



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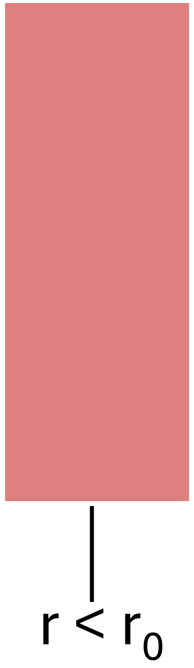
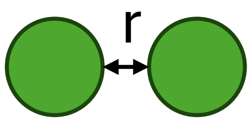
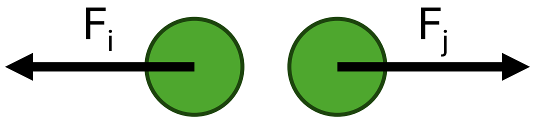




A repulsive force will push the two atoms apart towards the energy minimum



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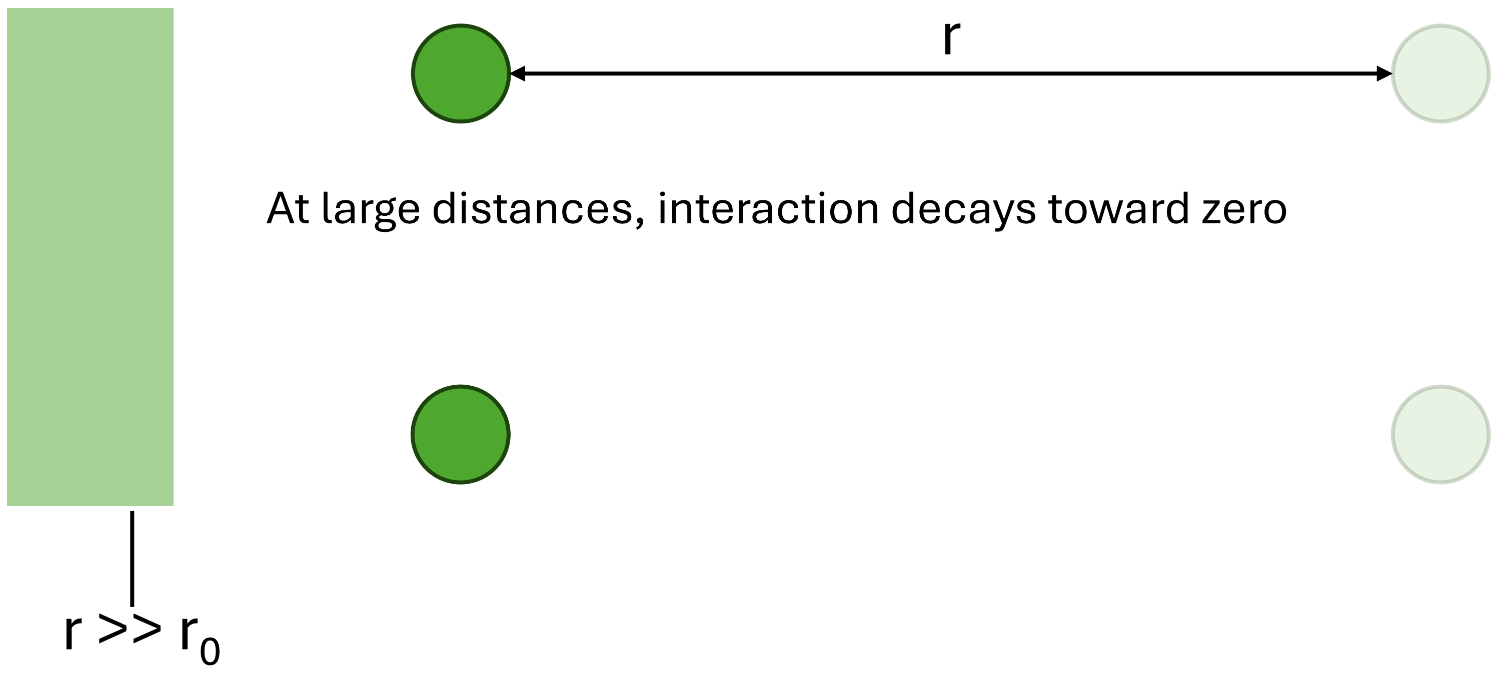

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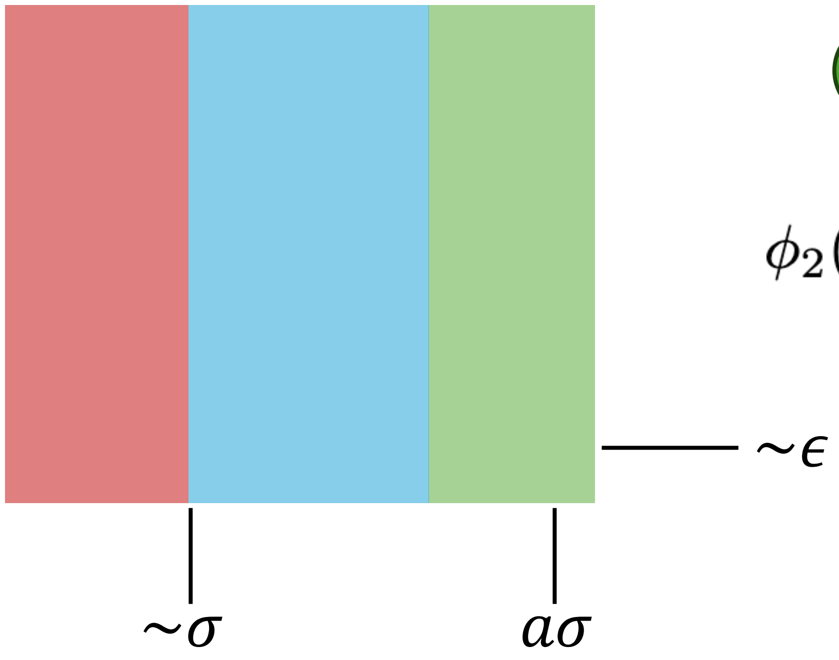
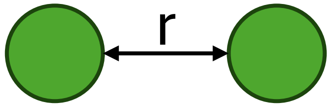


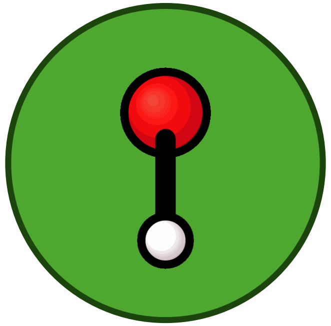


$$\phi_2(r_{ij}) = \epsilon A \left[B \left(\frac{\sigma}{r_{ij}} \right)^p - \left(\frac{\sigma}{r_{ij}} \right)^q \right] \exp\left(\frac{\sigma}{r_{ij} - a\sigma} \right)$$

BEYOND PAIR POTENTIALS

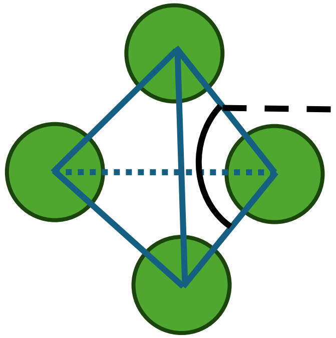
- To represent an entire water molecule as one bead we need some angles

Stillinger-Weber Angular Potential



$$\phi_3(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) = \epsilon\lambda \exp\left(\frac{\gamma\sigma}{r_{ij} - a\sigma}\right) \exp\left(\frac{\gamma\sigma}{r_{ik} - a\sigma}\right) \left(\cos\theta_{jik} + \frac{1}{3}\right)^2$$

This is not a spherically symmetric molecule!



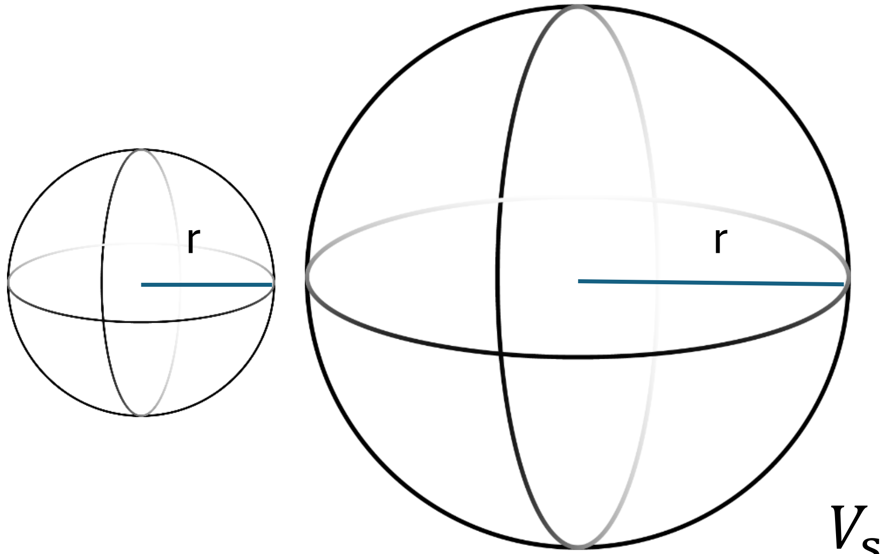
$$\cos(109.5^\circ) = -\frac{1}{3}$$

Our total potential energy term:

$$E_{\text{potential}} = \sum_{i<j} \phi_2(r_{ij}) + \sum_{i<j<k} \phi_3(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k)$$

A CRITICAL APPROXIMATION

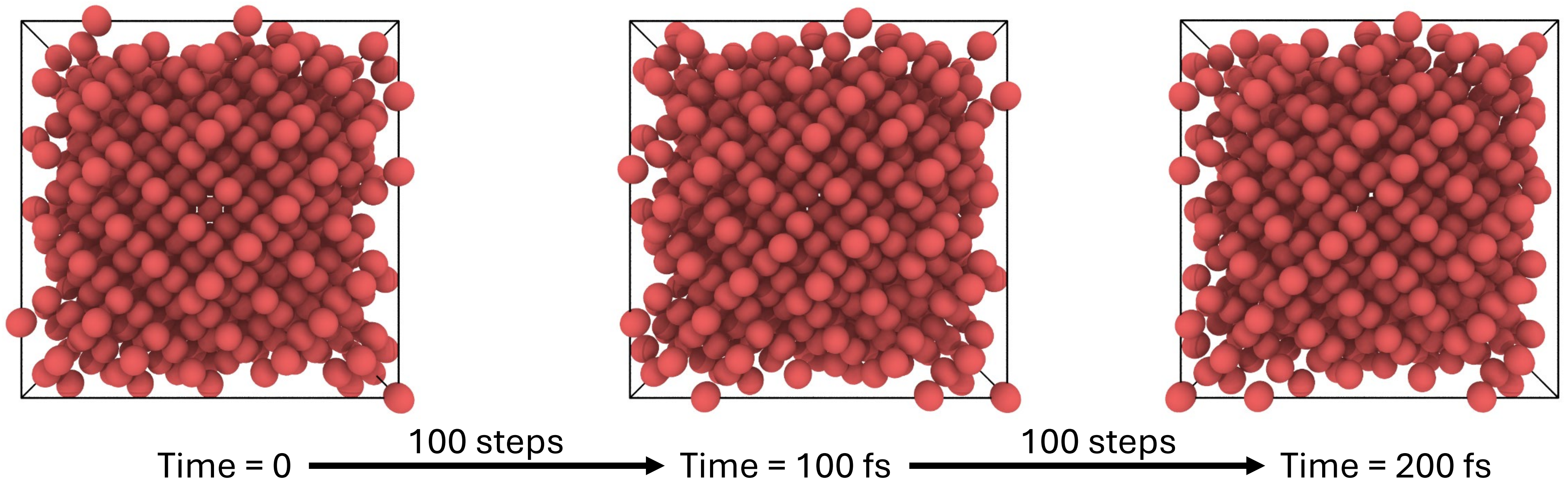
- Our potential energy decays towards zero as molecules separate
  - We can improve from $O(N^2)$ to $O(N)$ energy/force calculations when we can assume local interactions
  - The locality should be as strong as possible (small cutoff)



$$V_{\text{sphere}} = \frac{4}{3}\pi r^3$$

USING NEIGHBOR LISTS

- Each timestep calculation has time-correlated structure



- We can compute and cache neighbor lists to reduce $O(N^2)$ distance checks

USING NEIGHBOR LISTS

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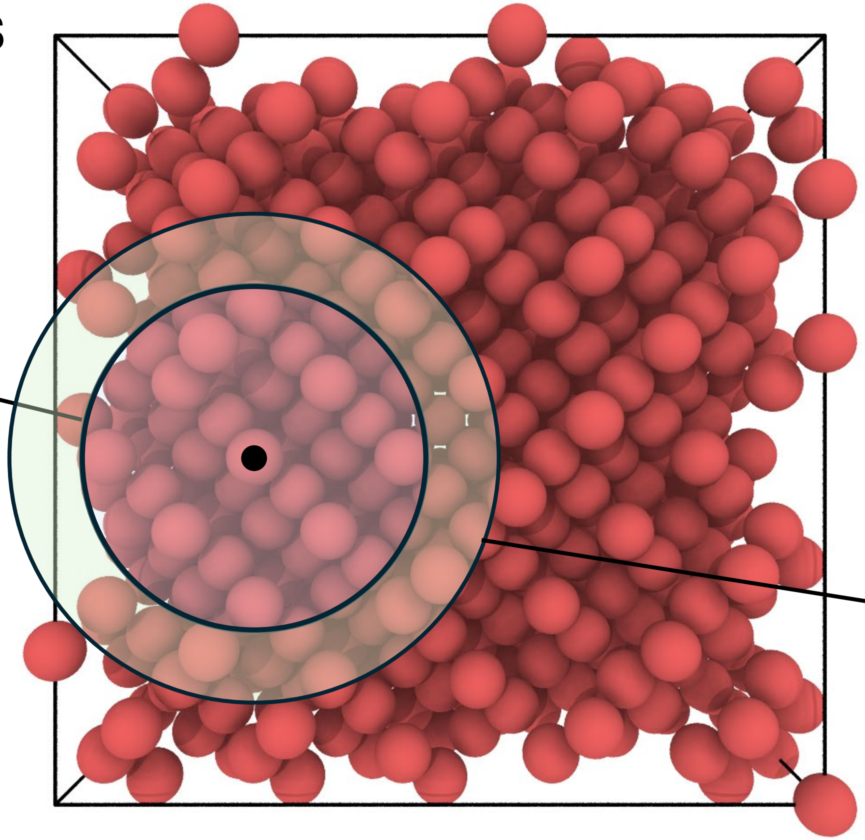
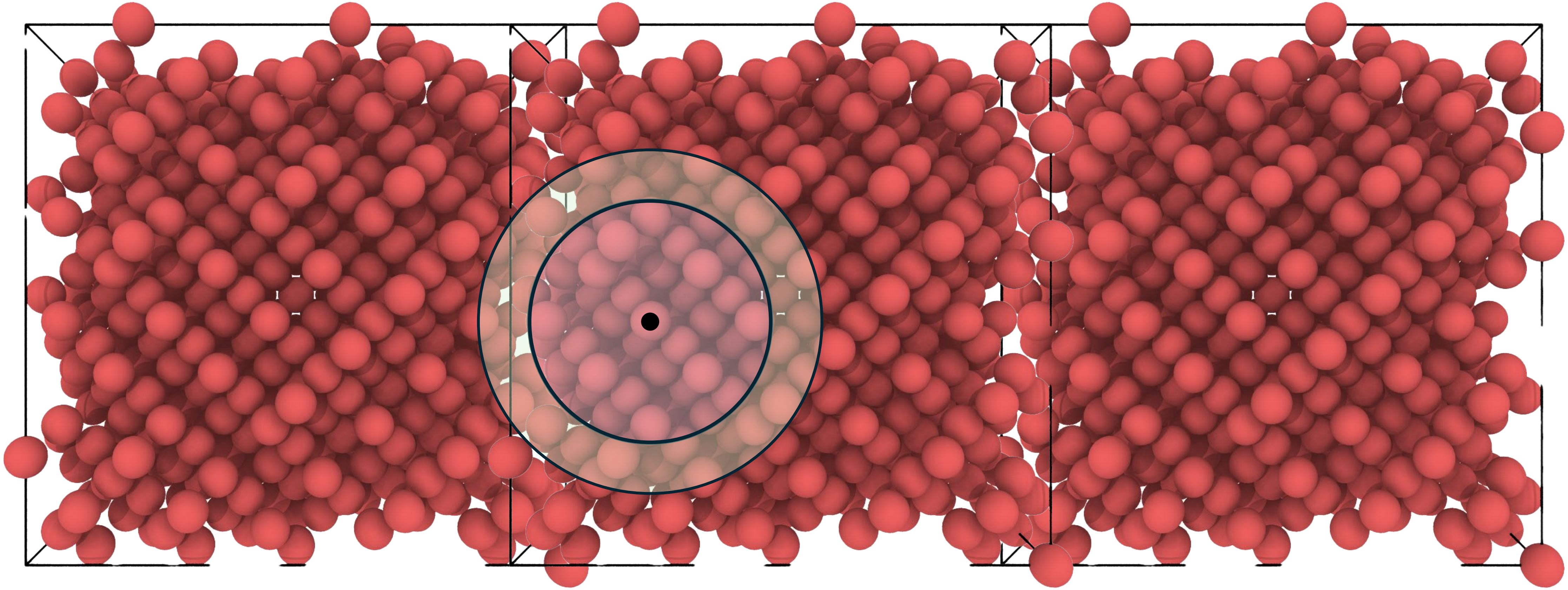


Atom	Neighbors
1	2, 3, 5, 37, 42 ...
2	1, 3, 11, 15, 37...

We now have fewer atoms to check each timestep!

PERIODIC BOUNDARIES

- We can approximate a much larger system than we actually compute if we make the system periodic



OK, LETS BEGIN!
CAN WE MELT ICE?

- Exercise 2

- Our system conserves energy but doesn’t quite do what we want.

THERMOSTATS

- We want to simulate a system at a particular environmental temperature condition (~Kinetic Energy)

$$E_{\text{total}} = E_{\text{kinetic}}(\mathbf{p}) + E_{\text{potential}}(\mathbf{r})$$

$$T = \frac{1}{3Nk_B}\sum_{i=1}^{N} m_i v_i^2$$

- But our integrator only preserves Total Energy

- To manage Kinetic Energy, we use a velocity-rescaling thermostat:

$$\mathbf{v}_i = \sqrt{1 + \frac{\Delta t}{\tau}\left(\frac{T_0}{T} - 1\right)}\,\mathbf{v}_i$$

NOW WE CAN CONTROL THE TEMPERATURE

- Exercise 3

- Our ~1-minute limit doesn't let us get very far...

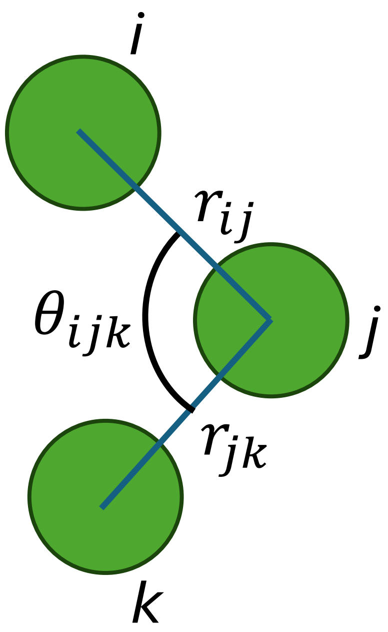
MOLECULAR DYNAMICS SIMULATION

Parallelization Techniques

SHARED MEMORY PARALLELISM

- Many operations in the MD Engine are loops over each atom:
  - State updates (position, velocity, etc.)
  - Force calculations over pairs, triplets of atoms
- We just need to handle race conditions:

$$F_j = \phi_2(r_{ij}) + \phi_2(r_{ji}) + \phi_3(r_i, r_j, r_k) + \cdots$$



POINTWISE UPDATES

- Integration Loop:

  For i in (all atoms):

  Update position array values

  Update velocity array values

- Thermostat Loop:

  For i in (all atoms):

  Apply velocity scaling factor

ENERGY/FORCE REDUCTIONS

- Energy Loop:

    For i in (all atoms):
        For j in (neighbor atoms):
            Compute pair, triplet energy
            E_i += energy

- Force Loop:

    For i in (all atoms):
        For j in (neighbor atoms):
            Compute pair, triplet force
            F_i += force

HOW FAR DOES THIS GET US?

- Exercise 4

- The water molecules are starting to wake up.

THURSDAY

- We will add some more complexity to the simulation algorithm
- We will melt our ice cube with MPI + CUDA accelerations